

Vibe Coding with Google Anti Gravity

Let's use AI tools to the fullest

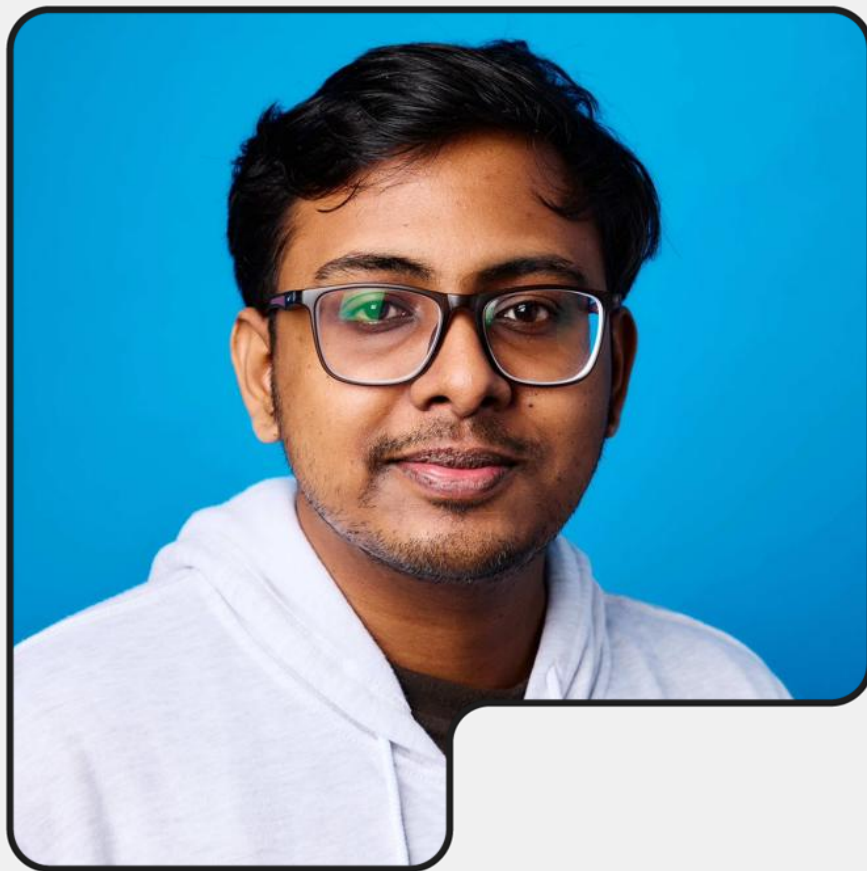


{ Build  with AI }

MD Shahriyar AI Mustakim Mitul

Lead @ML, AI, DL & NLP Community

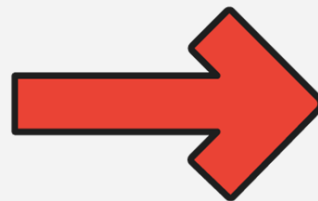
Awarded the Google Community
Champion Award [2024,2025]



Chapter

Join the ML, AI, DL & NLP Community

Join the group





**“No technology has
me dreaming bigger
than AI. It is the
biggest platform
shift of our lifetime.”**

- Sundar Pichai

CEO, Google



I think that most people are underestimating just how radical the upside of AI could be, just as I think most people are underestimating how bad the risks could.

- Dario Amodei





code_slides.txt

// A couple more notes on code slides:

// *When possible, use the “move in” (bottom to top) transition

// *Use the Google Sans Mono font

// *Set line spacing to 1.2

// *Don't use font sizes below 25

code_slides.txt

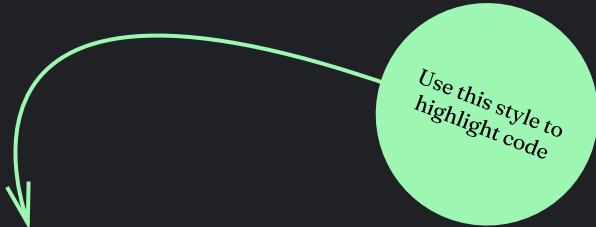
```
protected void onTryUpdate(int reason) throws RetryException {  
    // Do some awesome stuff  
    int foo = 15;  
    publishArtwork(new Artwork.Builder()  
        .title(photo.name)  
        .imageUri(Uri.parse(photo.image_url))  
        .newIntent(new Intent(Intent.ACTION_VIEW,  
            Uri.parse("http://500px.com/photo/" + photo.id)))  
        .build());  
    scheduleUpdate(System.currentTimeMillis() + ROTATE_TIME_MILLIS);  
}
```

Use this template
for code snippets
longer than
6 lines

Copy / paste your
code using **this tool**
for formatting using
the 'Dark Alternate'
theme

code_slides.txt

```
protected void onTryUpdate(int reason) throws RetryException {  
    // Do some awesome stuff  
    int foo = 15;  
    publishArtwork(new Artwork.Builder()  
        .title(photo.name)  
        .imageUri(Uri.parse(photo.image_url))  
        .viewIntent(new Intent(Intent.ACTION_VIEW,  
            Uri.parse("http://500px.com/photo/" + photo.id)))  
        .build());  
    scheduleUpdate(System.currentTimeMillis() + ROTATE_TIME_MILLIS);  
}
```



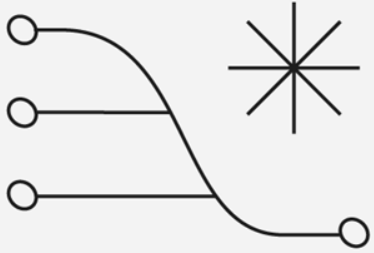
Use this style to
highlight code

code_slides.txt

```
protected void onTryUpdate(int reason) throws RetryException {  
    // Do some awesome stuff  
    int foo = 15;  
    publishArtwork(new Artwork.Builder()  
        .title(photo.name)  
        .imageUri(Uri.parse(photo.image_url))  
        .viewIntent(new Intent(Intent.ACTION_VIEW,  
            Uri.parse("http://500px.com/photo/" + photo.id)))  
        .build());  
    scheduleUpdate(System.currentTimeMillis() +  
ROTATE_TIME_MILLIS);  
}
```

Photo

Google for Developers



**Quote, short text, or
diagram goes here.**



{ Build  with AI }

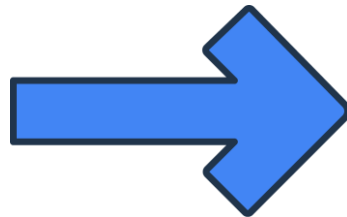
Vibe Coding with Google AI

Build your first app with Google AI Studio & Antigravity

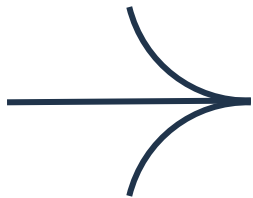
Get free access to this handbook

goo.gl/itsvibecoding

Updated 11 Apr 2026 | [Make a Copy](#)



What is Vibe Coding?



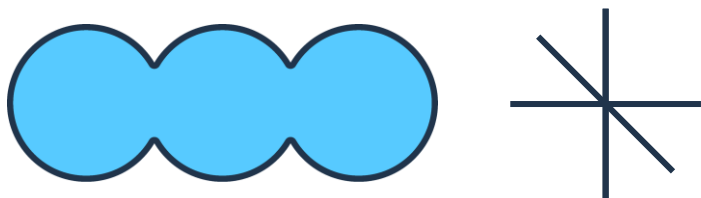
This term was coined by Andrej Karpathy in 2025

- Vibe Coding is a simple approach to build apps using natural language.
- You give instructions in simple language “Hey! build me an xx app that does yy” to an agentic platform and it builds the app for you.
- In most cases you can go from a simple idea to a live app preview in ideally less than 15 mins.
- **No programming or coding experience required.**

Why “Vibe Coding” is for you

THE TRADITIONAL WAY

- Learning complex syntax and languages
- Months of development of a prototype
- Getting stuck on a single bug or error.

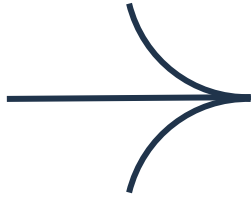


THE VIBE CODING WAY

- Describing your idea in plain English.
- Building fully functional apps in minutes.
- Clicking "Fix" and letting the AI self-heal the code.

This is arguably the best time in history to learn how to build.

People are building their own apps with vibe coding to showcase their craft & creativity and to bring new ideas to life.



The rise of 'micro' apps: non-developers are writing apps instead of buying them



Business Insider

<https://www.businessinsider.com> › Careers



Vibe coding is becoming a real job now

Google offers the full-stack for vibe coding. Follow 3 simple steps to take your app live

Step 1: Think

Think (The Intent)

Define the core problem you want to solve. Whether it's a productivity utility or a retro game, focus on the intent you'll describe to the AI

Step 2: Build

Build (The Vibe)

Use AI Studio for instant prototypes or Antigravity for agentic coding. Describe your idea and watch the agent build the logic and UI in real-time

Step 3: Publish

Publish (The Launch)

Deploy your app instantly to Google Cloud. Go from a local preview to a public URL that you can share with your team, users, or the world



Stitch



Gemini



AI Studio



Antigravity



Google Cloud Run

Google's vibe coding tools & platforms



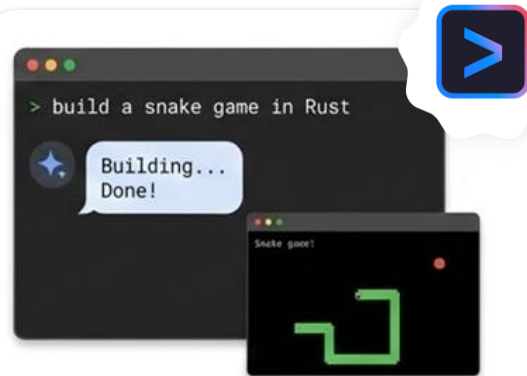
AI Studio: Rapid Prototyping

For **Builders & Curious Creators** who want to describe or draw an app and get a functional UI with **live preview** instantly.



Antigravity: Agentic IDE

For **Adv Builders and Developers** who are comfortable in an IDE environment but want an **AI "partner"** to do the heavy lifting.



Gemini CLI: Smart Terminal

For **Terminal Purists and Developers** who prefer text-based, high-speed workflow for rapid iteration and **system-level tasks**.

Finding your entry point in vibe coding

Tool	Starting Point	Skill Level	Coding approach	Key feature
Google AI Studio	An idea you want to see, fast	Beginner. No coding experience needed	No code / Low Code	Single-prompt app generation with zero friction deployment
Google Antigravity	An engineering task or mission	Beginner to advanced	Agent-first / Autonomous	Mission Control for orchestrating autonomous agents across the editor, terminal, and browser.
Gemini CLI	Terminal based development	Intermediate to advanced	Terminal first / AI-assisted	Open-source agent for terminal-first "vibe" workflows

The Vibe Coder's Glossary

VIBE CODING

Orchestrating intent, not syntax

You can open Gemini and type: "Build me a to-do app with dark theme, satisfying check-off animation, and local storage."

You never wrote a single line of HTML, CSS, or JS. Gemini infers the stack, writes the code, and produces a working app

MODEL CONTEXT PROTOCOL

The "universal connector"

You connect Gemini to your GitHub repo via an MCP server. Now you can ask: "**What changed in the last 3 pull requests?**" – and Gemini actually reads your real repo, not a fictional one.

AGENTIC AI

AI that acts, not just answers

You tell Gemini Code: "Fix all the broken tests in my project."

It reads your test files, runs **npm test** in the terminal, sees which tests fail, edits the source files, re-runs tests all on its own, without you clicking anything.

CONTEXT WINDOW

The AI's "active memory."

Assume you are 40 messages deep into debugging a feature. With a small context window, the AI has already "forgotten" your original requirements from message 1.

With a large one (200k + tokens), it still remembers the architecture decisions you made an hour ago.

TOKENS & QUOTA

The AI's unit of fuel

You paste your 800-line codebase and ask Gemini to refactor it. That might cost around 12,000 tokens (input + output). If your plan allows 100,000 tokens/day, that's 12% of your daily quota gone in one task.

ARTIFACTS

The concrete outputs AI creates

You ask Gemini to design a landing page. Instead of just showing you code in chat, it produces an artifact – a live preview panel you can interact with right in the conversation. That rendered page is the artifact. So is a generated diagram, a downloadable CSV, or a saved Python script.

Building simple apps with vibe coding



Retro Arcade Games



Interactive Dashboards



Productivity Utilities



Personal Portfolios



Custom Media Players



Smart Calculators



Interactive Storyboards



Resource Libraries



Event Countdowners



Daily Habit Logs

AI features you can integrate in your app



Nano Banana Powered app

Add powerful photo editing to your app. Allow users to add objectives, remove backgrounds or change a photo's style just by typing.



Analyze images

Enable your app to see and understand images. Allow users to upload a photo of a receipt, a menu or a chart to get instant data extraction, translation or summary.



Generate Speech

Give your app a voice. Add text-to-speech to read articles aloud, provide audio navigation, or create voice-based assistants for your users.



Gemini intelligence

Embed Gemini in your app to complete all sorts of tasks - analyze content, make edits and more.



Transcribe audio

Add a feature to provide live, real-time transcription of any audio feed for your users.



Fast AI responses

Add lightning-fast real time responses to your app using Flash Lite. Perfect for instant auto completes, or conversations agents that feel alive,



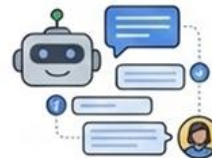
Use Google Search data

Connect your app to real-time Google Search results. Build an agent that can discuss current events, cite recent news or fact check information.



Prompt based video generation

Add video generation to your creative app. Let users turn their blog posts, scripts, or product descriptions into short video clips.



AI powered chatbot

Add a context-aware chatbot to your app. Give your users a support agent that remembers the conversation, perfect for multi-step booking and troubleshooting.



Use Google Maps data

Connect your app to real-time Google Maps data. Build an agent that can pull information about places, routes and directions.

The anatomy of a “Powerful” prompt

Expand your prompt into multiple layers

[Give a role]

[Define the goal]

[Provide context]

[Does it need AI?]

[Create the vibe]

Optional : [Add a visual]



Good prompt

INSTEAD OF THIS

Help me build an app to track my food and water intake.

Powerful prompt

TRY THIS

You're an app builder who prioritizes good user experience. Create a web app that helps me track my food calories and water intake. I plan to use this app daily to count my calories and track my water intake and I want weekly summary. It would be good to use AI to track my calories if I can provide pictures of my food. Create a skeuomorphic vibe. Use the attached mockup for inspiration.

Advanced prompting techniques



Iterate with Action History

Don't just look at the result; review the Action History to see exactly which files Gemini edited. If a specific logic or style is off, use the Live Preview to test the change instantly and ask for a targeted fix



The Layered Build Approach

Build your app in cycles rather than all at once. Start with the core functionality (the "logic"), then layer on the Aesthetic UI, and finally integrate your chosen AI Features like Nano Banana or Google Maps data



The Architect Interview

Before starting a complex project, ask Gemini: "I want to build [App Name]. Ask me 5 technical questions about my user flow and tech stack before you start coding". This ensures the Built result matches your vision perfectly



Blueprint Optimization

Ask Gemini to *"Rewrite my prompt into a professional technical specification"* before you run it. This helps ensure the code it generates is modular and follows best practices

Shifting to an era of action



Models like Gemini 3.0 have set the stage for vibe coding by going from mere understanding and thinking to agentic reasoning. This “agentic reasoning” capability allows vibe coding platforms to think, build and publish.

[Jump to AI Studio](#)

[Jump to Antigravity](#)

Getting started with Google Antigravity

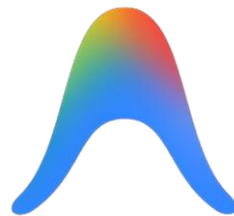
Antigravity is an agent-first tool. Its locally downloadable tool that allows high-speed workflows and deep access to your project files.

Switching between tools comes down to your preference: Use AI Studio for visual live previews, and move to Antigravity when you are ready for system-level implementation.

Switch from a single-prompt interface to a "Mission Control" where you can orchestrate autonomous agents across your editor, terminal, and browser simultaneously.

Antigravity

If you haven't already, please **install Antigravity** to follow along with the hands-on sessions later and build wonderful projects!



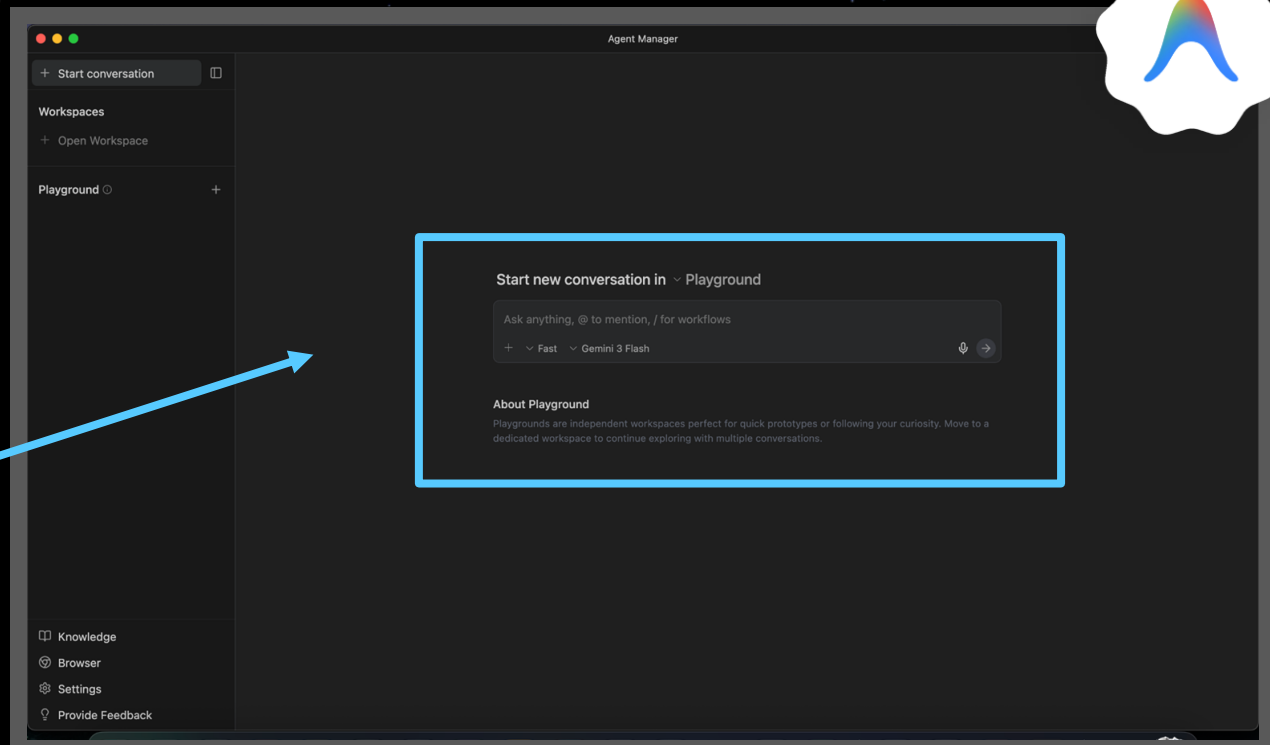
antigravity.google/download



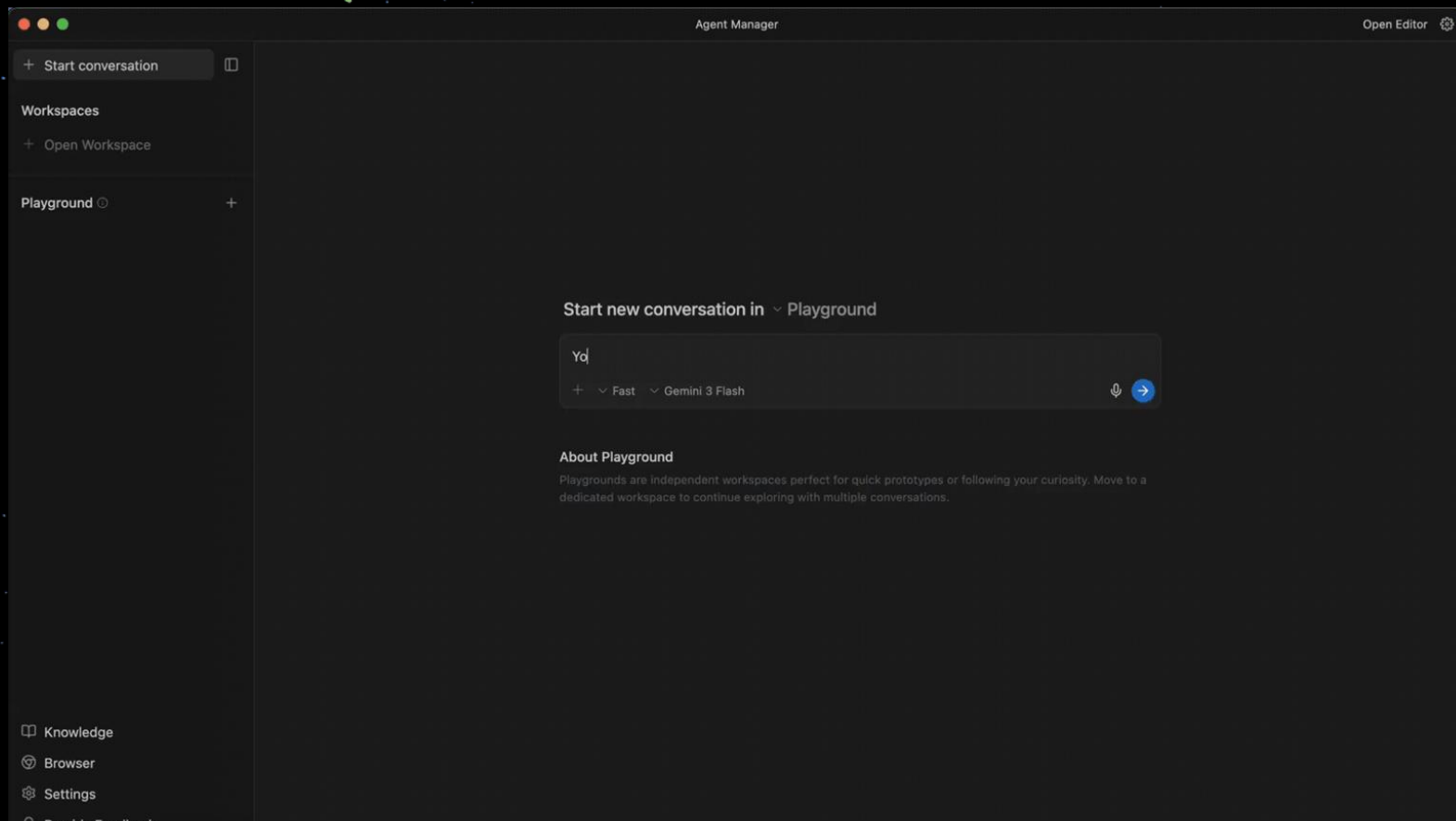
Build your first app in Agent Manager

Open Antigravity and enter your first app prompt

You're an app builder who prioritizes good user experience. Create a web app that helps me track my food calories and water intake. I plan to use this app daily to count my calories and track my water intake and I want weekly summary. It would be good to use AI to track my calories if I can provide pictures of my food. Create a skeuomorphic vibe.



Go from Prompt to Publishing in 15-minutes



[+ Start conversation](#)

Workspaces

[+ Open Workspace](#)

Playground

Agent Manager: For first-time vibe coders this is the place to start. You can start by giving a new prompt in the Playground

Start new conversation in Playground

Ask anything, @ to mention, / for workflows



Fast



Gemini 3 Flash



About Playground

Playgrounds are independent workspaces perfect for quick prototypes or following your curiosity. Move to a dedicated workspace to continue exploring with multiple conversations.

Knowledge

Browser

Settings

Provide Feedback

[+ Start conversation](#)

Workspaces

[+ Open Workspace](#)[Playground](#)

Start new conversation in Playground

Ask anything, @ to mention, / for workflows

+ Fast Gemini 3 Flash



About

Playground

Dedicated

Conversational Model

Planning

Agent can

research,

Fast

Agent will

that can b

Gemini 3.1 Pro (High) New

Gemini 3.1 Pro (Low) New

Gemini 3 Flash

Claude Sonnet 4.6 (Thinking)

Claude Opus 4.6 (Thinking)

GPT-OSS 120B (Medium)

Conversation Mode & Model This is where you can select between “Planning” and “Fast” as your conversation mode. Keep in mind that Planning mode consumes more tokens. Here you can also select your Model.

[Knowledge](#)[Browser](#)[Settings](#)[Provide Feedback](#)

 + Start conversation

Workspaces

+ Open Workspace

Playground 

+

Start new conversation in  Playground

Ask anything, @ to mention, / for workflows

+

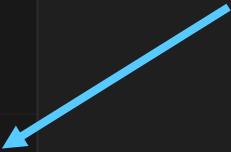
▼ Fast

▼ Gemini 3 Flash



About Playground

Playgrounds are independent workspaces perfect for quick prototypes or following your curiosity. Move to a dedicated workspace to continue exploring with multiple conversations.

 Knowledge Browser Settings Provide Feedback

Agentic Browser : Antigravity has a unique Agentic Browser mode which can launch Chrome and manage to complete tasks just like a human.

+ Start conversation

Workspaces

+ Open Workspace

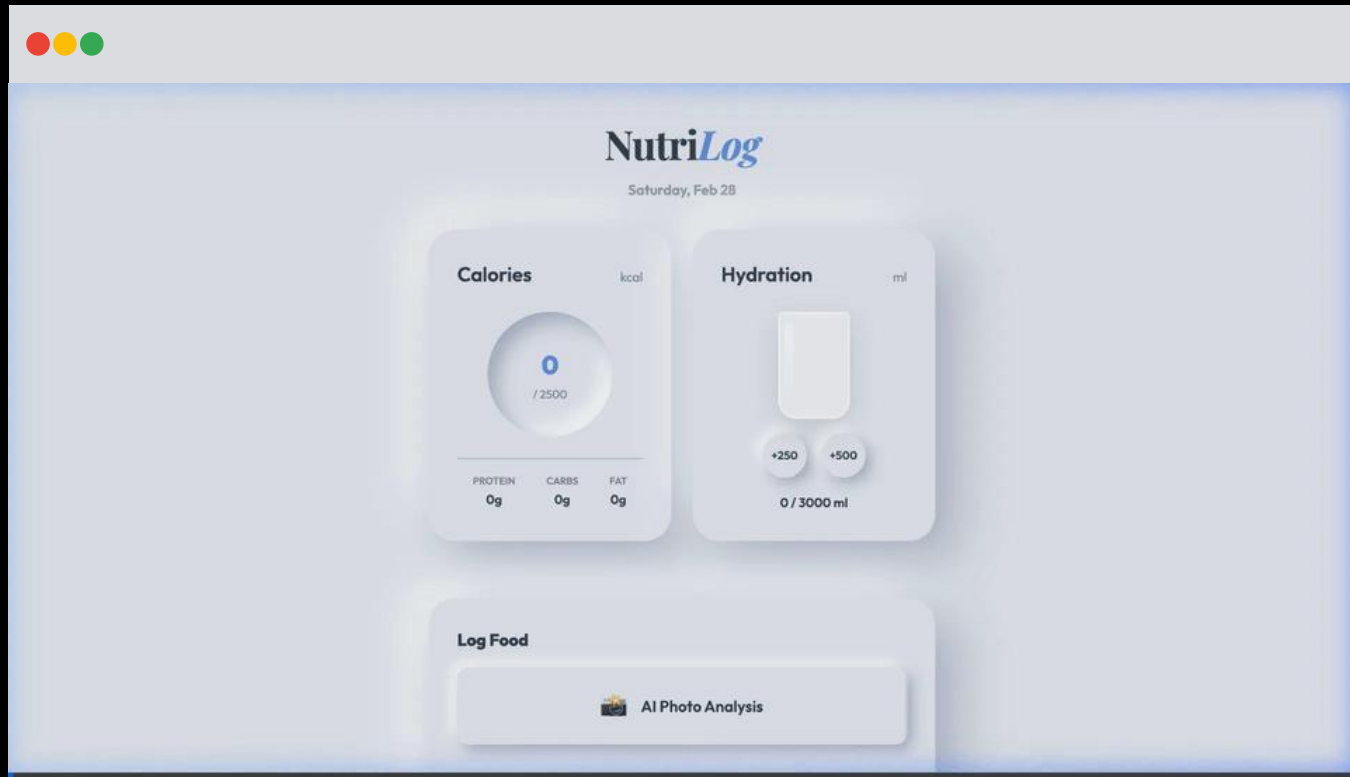
Playground ⓘ +

Knowledge

Browser

Settings

Provide Feedback



Agentic Browser : Antigravity's native integration with Chrome allows you to demo your app in full agentic mode inside your browser



+ Start conversation



Workspaces

+ Open Workspace

Playground ⓘ



Editor Mode : Once you have run the first iteration of your app you can switch to the Editor mode to see the files, artefacts, terminal etc



Start new conversation in ▾ Playground

Ask anything, @ to mention, / for workflows



▾ Fast

▾ Gemini 3 Flash



About Playground

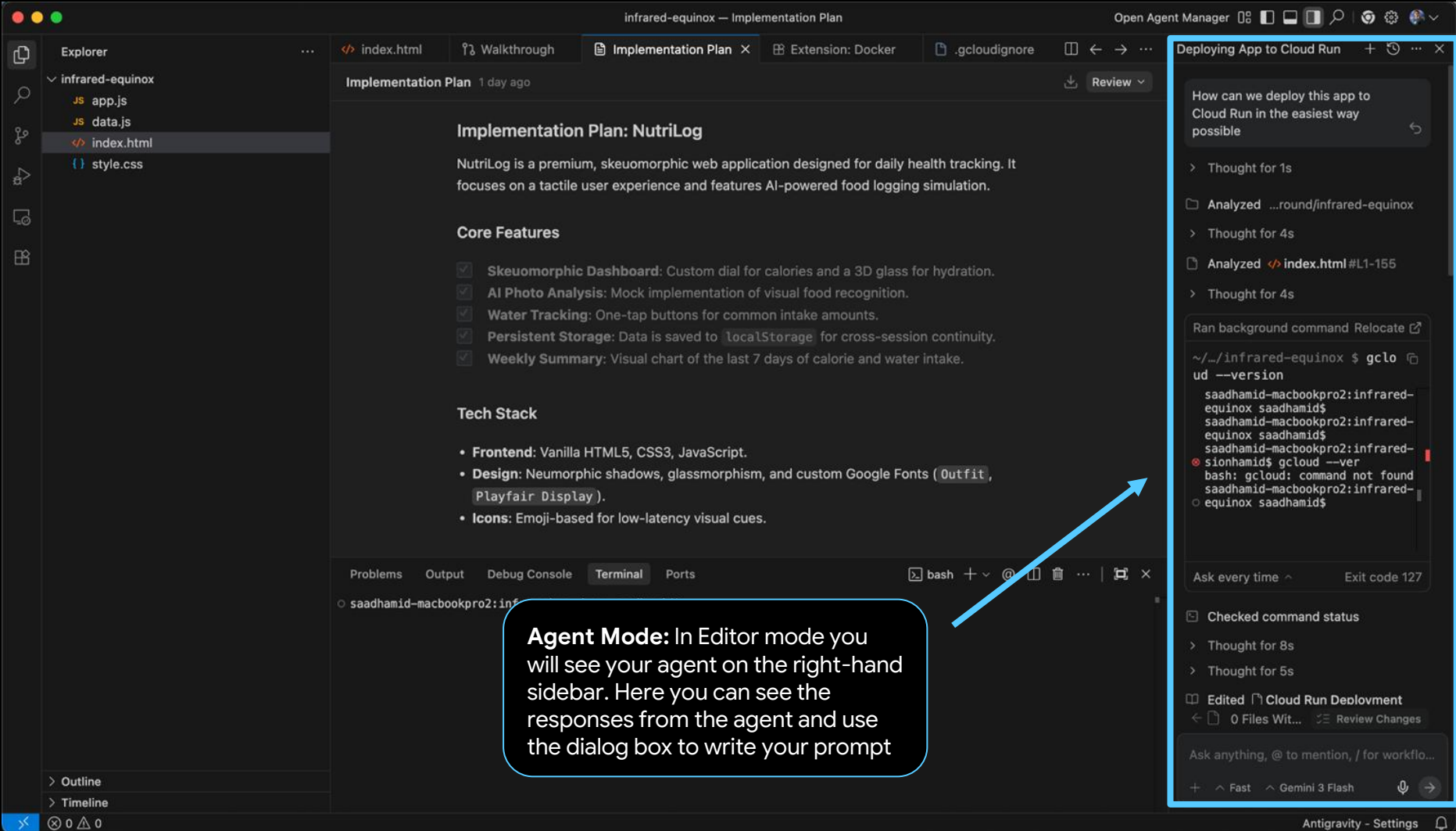
Playgrounds are independent workspaces perfect for quick prototypes or following your curiosity. Move to a dedicated workspace to continue exploring with multiple conversations.

Knowledge

Browser

Settings

Provide Feedback



Implementation Plan: NutriLog

NutriLog is a premium, skeuomorphic web application designed for daily health tracking. It focuses on a tactile user experience and features AI-powered food logging simulation.

Core Features

- ☒ **Skeuomorphic Dashboard:** Custom dial for calories and a 3D glass for hydration.
- ☒ **AI Photo Analysis:** Mock implementation of visual food recognition.
- ☒ **Water Tracking:** One-tap buttons for common intake amounts.
- ☒ **Persistent Storage:** Data is saved to `localStorage` for cross-session continuity.
- ☒ **Weekly Summary:** Visual chart of the last 7 days of calorie and water intake.

Tech Stack

- **Frontend:** Vanilla HTML5, CSS3, JavaScript.
- **Design:** Neumorphic shadows, glassmorphism, and custom Google Fonts (`Outfit`, `Playfair Display`).
- **Icons:** Emoji-based for low-latency visual cues.

saadhamid-macbookpro2:inf

Agent Mode: In Editor mode you will see your agent on the right-hand sidebar. Here you can see the responses from the agent and use the dialog box to write your prompt

How can we deploy this app to Cloud Run in the easiest way possible

- > Thought for 1s
- 📁 Analyzed ...round/infrared-equinox
- > Thought for 4s
- 📄 Analyzed <code>index.html</code> #L1-155
- > Thought for 4s

Ran background command Relocate

```
~/~/infrared-equinox $ gclo  
ud --version  
saadhamid-macbookpro2:infrared-  
equinox saadhamid$  
saadhamid-macbookpro2:infrared-  
equinox saadhamid$  
saadhamid-macbookpro2:infrared-  
equinox saadhamid$ gcloud --ver  
bash: gcloud: command not found  
saadhamid-macbookpro2:infrared-  
equinox saadhamid$
```

Ask every time ^ Exit code 127

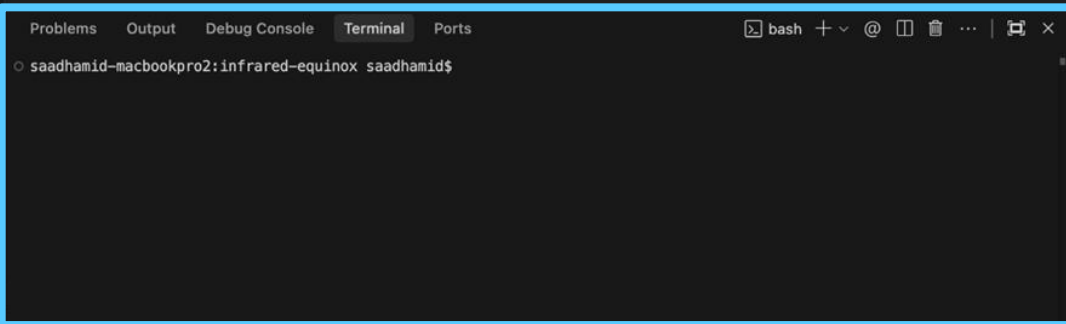
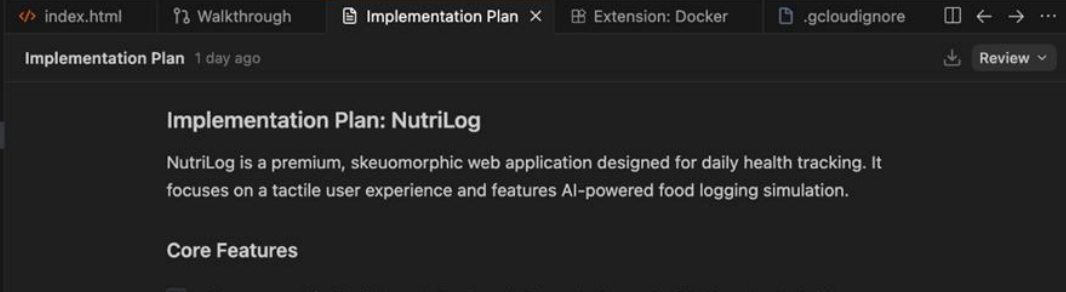
- 📄 Checked command status
- > Thought for 8s
- > Thought for 5s

📄 Edited Cloud Run Deployment

< 0 Files Wit... Review Changes

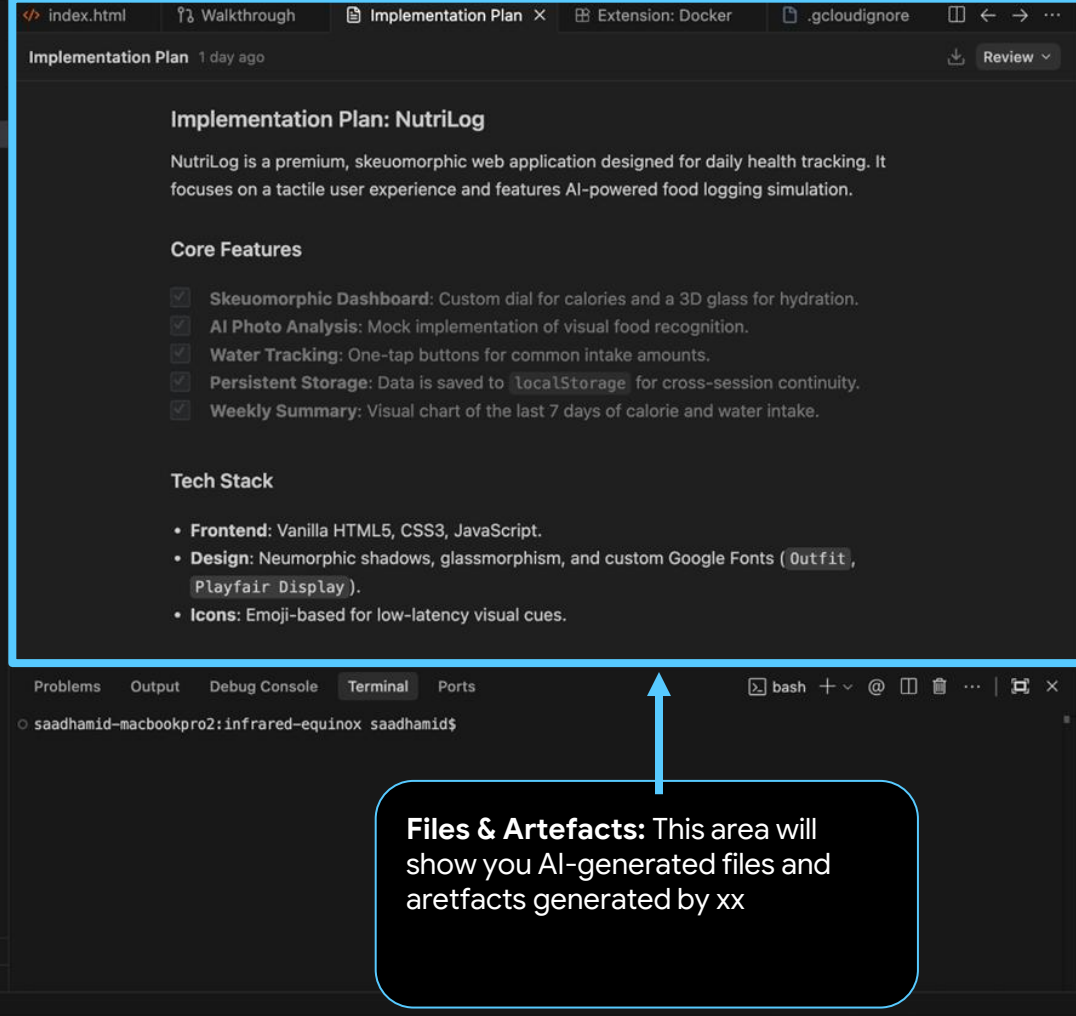
Ask anything, @ to mention, / for workflo...

+ ^ Fast ^ Gemini 3 Flash



```
> Thought for 1s
  Analyzed ...round/infrared-equinox
> Thought for 4s
  Analyzed < index.html #L1-155
> Thought for 4s
```

Ask every time ^ Exit code 12



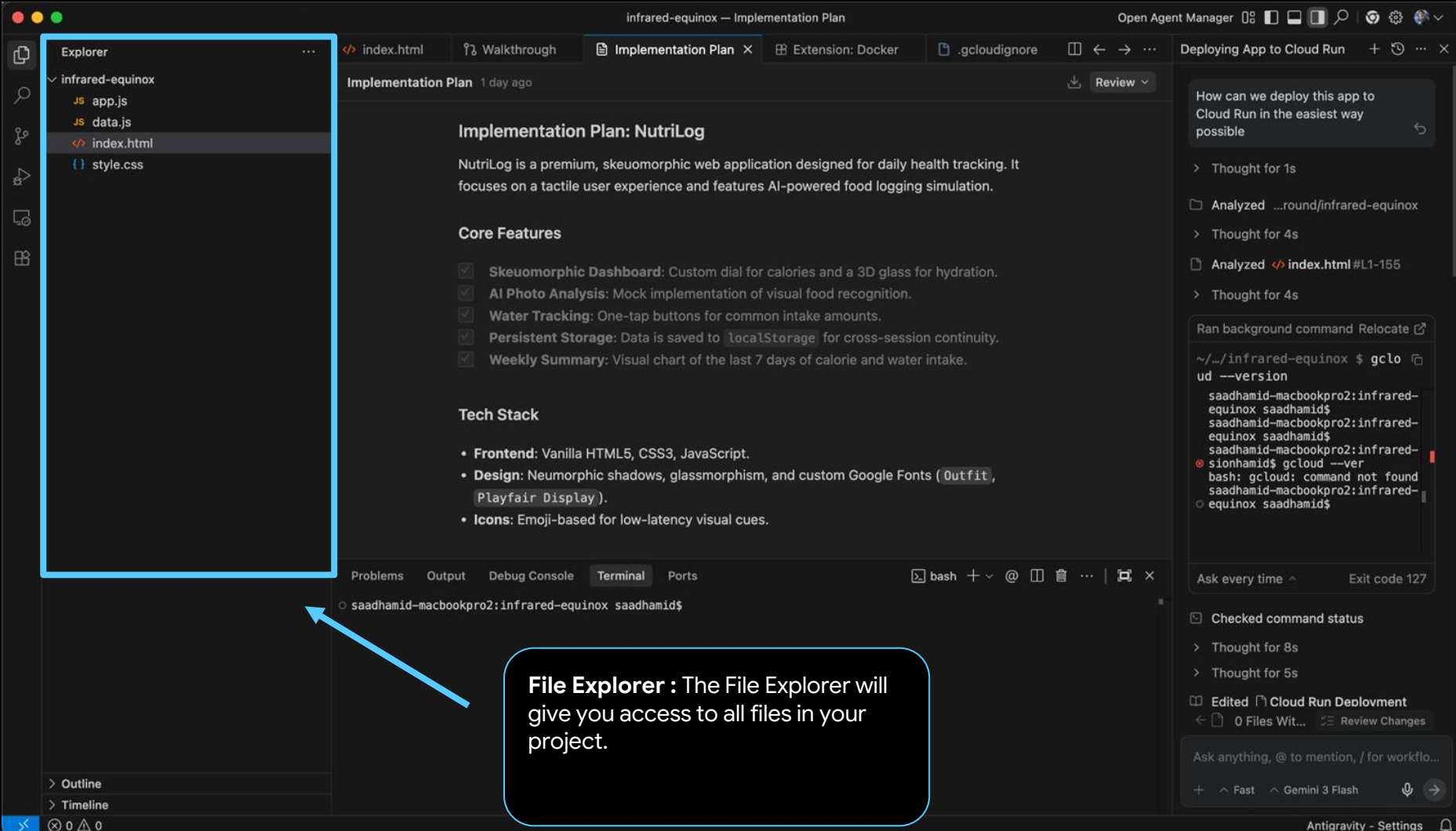
Files & Artefacts: This area will show you AI-generated files and artefacts generated by xx

- > Thought for 1s
- 📁 Analyzed ...round/infrared-equinox
- > Thought for 4s
- 📁 Analyzed 🔗 index.html #L1-155
- > Thought for 4s

Ask every time ^ Exit code 127

Edited Cloud Run Deployment
0 Files Wit... Review Changes

+ ^ Fast ^ Gemini 3 Flash

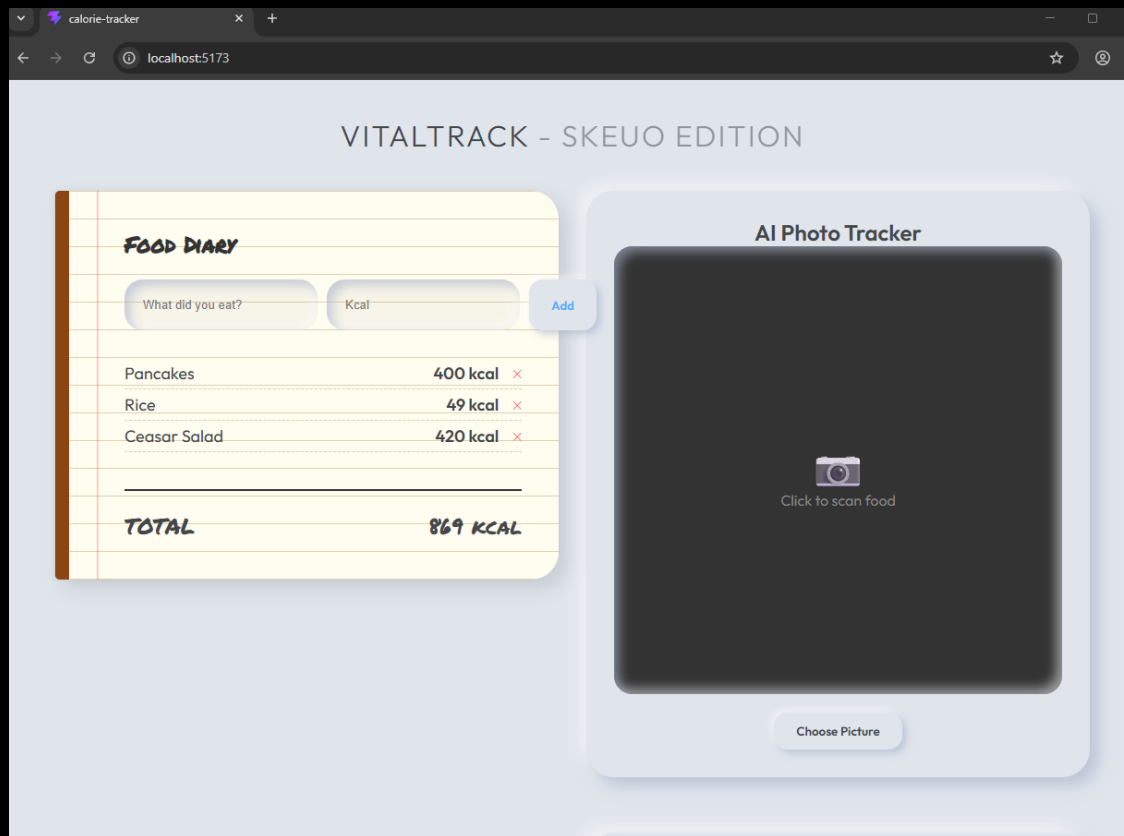


File Explorer : The File Explorer will give you access to all files in your project.

Prompting techniques to keep in mind

- Define the "What" (Goal), the "How" (Constraints), and the "Success Criteria" (Definition of Done).
- Assigning a specific seniority or specialty to an agent changes its behavior and the tools it prioritizes.
- Use @filename or @directory to point the agent exactly where to look.
- After implementation, run the app and record a browser session video showing the successful login flow
- If you have proprietary logic (e.g., a custom internal API), don't explain it in every prompt. Instead, create a SKILL.md file in the .agent/skills/ directory.
- Please perform a security audit on this project specifically for API keys. Check if any keys are hardcoded in the source. If they are, refactor the code to use environment variables via a .env file. Ensure the .gitignore is properly configured to block .env and any other sensitive files from being tracked by Git.

The app might look like this



Get Started with Cloud Credits

Request resources to deploy and share your vibe-coded apps

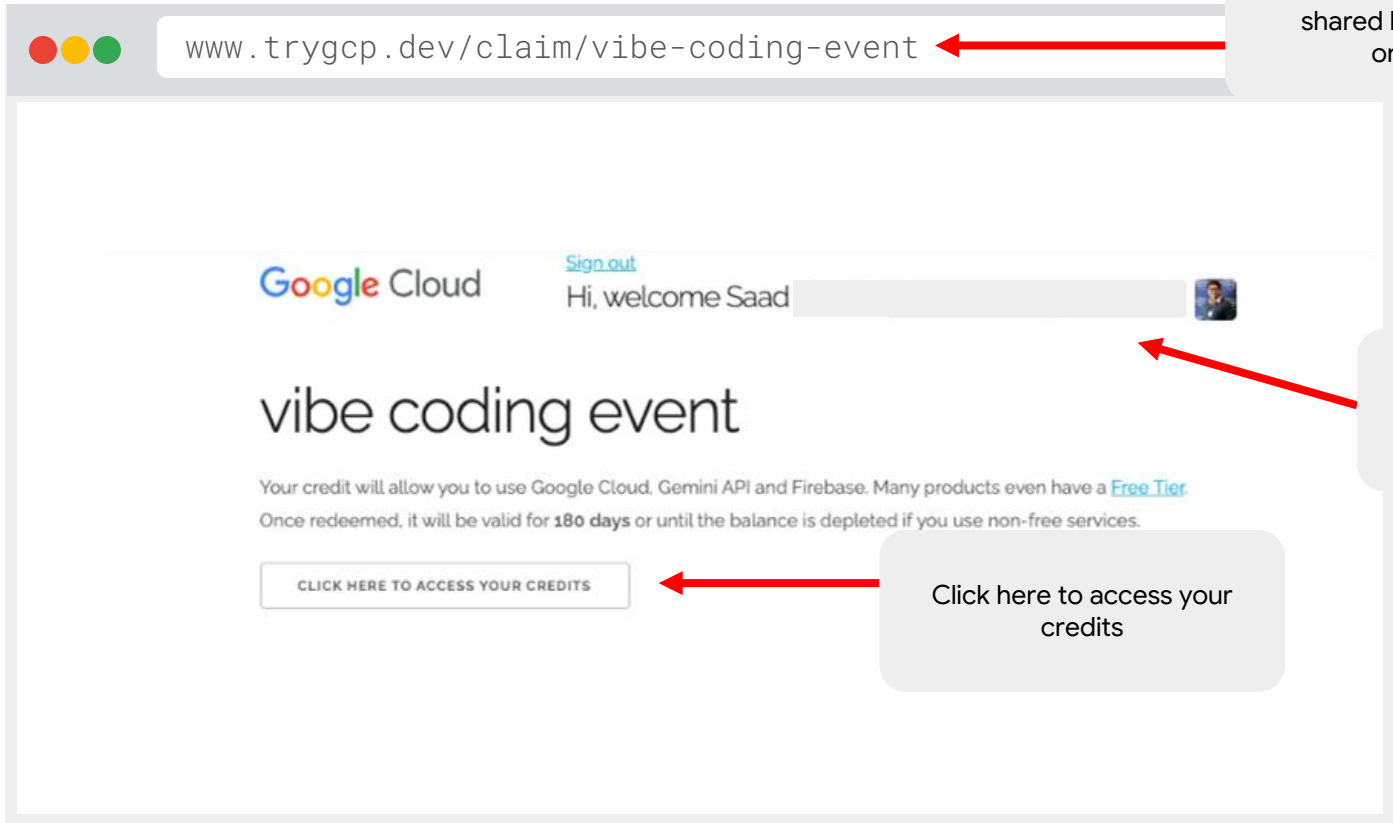
goo.gle/community-cloud-request

Why request credits?

Deploy and publish your apps instantly to the world. Credits cover your hosting needs.

Already have a link?

If you've received your credit link, proceed to the next step to redeem and start building.



www.trygcp.dev/claim/vibe-coding-event

Enter credit claim URL
shared by your event
organizer

Google Cloud

[Sign out](#)

Hi, welcome Saad



vibe coding event

Your credit will allow you to use Google Cloud, Gemini API and Firebase. Many products even have a [Free Tier](#).
Once redeemed, it will be valid for **180 days** or until the balance is depleted if you use non-free services.

CLICK HERE TO ACCESS YOUR CREDITS

Click here to access your
credits

Make sure you are
signed in with your
Personal Gmail



console.cloud.google.com

Google Cloud

Select a project

GCP credit application

Fill in the following information below to apply GCP credits to your account listed below.

First name *
Vibe

Last name *
Coder

Account email
saadh.work@gmail.com

Credits will be applied to this account. If you'd like to apply credits to a different account, specify your preference [here](#).

Coupon code
1UHT-BNN1-06ER-NA32

Terms and conditions

The following terms and conditions apply to the credit you received for Google Cloud products (the "Credit(s)").

Accept and continue

* Indicates required

Accept the terms and
continue



console.cloud.google.com

Google Cloud

Billing / Overview

Billing account
Google Cloud Platform Trial Billing

Overview

Paid account

Manage billing account

Overview

Cost management

Reports

Cost table

Cost breakdown

Budgets & alerts

Billing export

Anomalies

Cost optimization

FinOps hub

Committed use discounts

CUD analysis

Pricing

Cost estimation

Credits

Billing management

Account management

Your total cost (February 1 – 22, 2026)

Cost: \$0.00

Savings: \$0.00

Total cost: \$0.00

Forecasted total cost

Available once there is enough usage

Costs take a few hours to show up, and might take longer than 24 hours.

Learn more

View details on Reports

Credit successfully applied

You can edit your billing account name here

Notice you are on Google Cloud Trial Billing

Click on "Credits" link in the sidebar menu to verify credits

You will see a "Credit successfully applied" popup note.



console.cloud.google.com

Google Cloud

Search (/) for resources, docs, products, and more

Search



Billing / Credits / All Credits

Billing account
TryGCP Feb Credit

Credits

Overview

Cost management

Reports

Cost table

Cost breakdown

Budgets & alerts

Billing export

Anomalies

Cost optimization

FinOps hub

Committed use discounts (CU...)

CUD analysis

Pricing

Cost estimation

Credits

Billing management

Account management

Issued Credits

View and download credit details here. Active committed use discounts are not included here and can be viewed on the [Commitments page](#).

Filter Filter credits

Credit name	Status ↑	Percent remaining	Remaining value	Original value	Type	Credit ID	Usage scope ⓘ	Start date	End date	Credit
TryGCP - Instrumentless credit	Available	100%	\$5.00	\$5.00	One-time	QPYP41GG...	Any service on this billing account.	February 22, 2026	August 21, 2026	Net pri

If you renamed your trial billing account with free credits, you will see the new name here.

In this row you will see value of the credits, including start date and end-date and % remaining.

Using Cloud Credits for Vibe Coding

Vibe Coding on AI Studio

Your AI Studio model quota is linked to free tier (unless you have purchased more quota). The cloud credits provided cannot be used for model usage however you can use the cloud credits to deploy your vibe-coded app to Cloud Run for free.

[Follow the guide](#)

Vibe Coding on AntigraVity

Antigravity is **free** and your AntigraVity mo**Google AI subscription**del quota is linked to your. However, you can use the **cloud credits provided to deploy your app to Google Cloud Run** via the instructions provided.

[Follow the guide](#)

Vibe Coding on Gemini CLI

Free tier users can use flash models in Gemini CLI. You can follow either of these options to unleash pro models or use cloud run. 1) user Vertex AI authentication to unleash the pro models. 2) guide them deploy the vibe-coded app to Cloud Run.

[Follow the guide](#)

Once done, the app will be launched in the Cloud

The **VitalTrack** app is officially live! 🚀

I have successfully built, verified, and deployed your skeuomorphic calorie and water tracker to Google Cloud Run.

🌟 Live Application:

<https://calorie-tracker-830998306846.asia-east1.run.app>

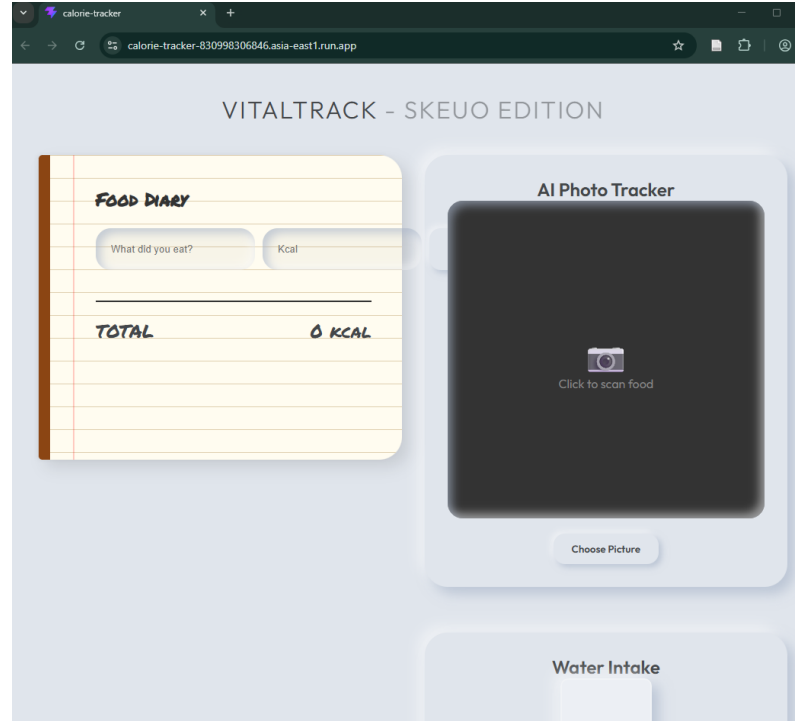
Project Status:

- **Design:** Premium skeuomorphic aesthetic with a physical food diary and filling water glass.
- **Functionality:** Local storage persistence, AI scanner simulation, and weekly progress charts are all active.
- **Infrastructure:** Hosted on Cloud Run in `asia-east1` within project `buildwithai2026-493112`.

You can view the full verification details and final screenshots in the 📄 [walkthrough.md](#).
Enjoy your tracking!

Files Modified 6

🔗 [App.tsx](#) 🔗 [WaterTracker.tsx](#) 🔗 [FoodNotebook.tsx](#) 🔗 [AICamera.tsx](#) 🔗 [Weekly.tsx](#)



Thank you!

Happy Building #BuildwithAI

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